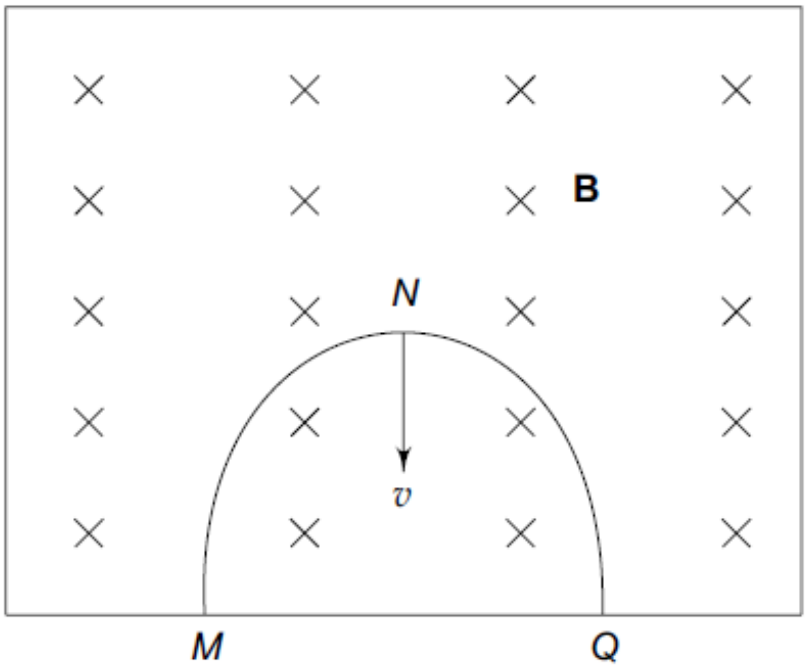
# Quizrr Chapter-Wise Test for JEE Main - 2025

**Q.1**

A thin semicircular conducting ring of radius $R$ is falling with its plane vertical in a horizontal magnetic field $B$ (Fig.). At the position $MNQ$ the speed of the ring is $v$ and the potential difference across the ring is




**(A)** zero

**(B)** $\frac{1}{2}Bv\pi R^2$ and $M$ is at higher potential

**(C)** $\pi RBv$ and $Q$ is at higher potential

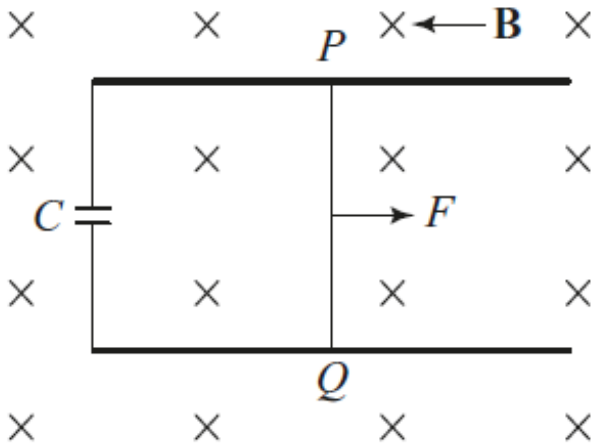
**(D)** $2RBv$ and $Q$ is at higher potential.

**Q.2**

A wire $PQ$ of mass $m = 10\ \text{g}$ and length $l = 25\ \text{cm}$ can freely slide on horizontal, smooth and parallel rails placed in a uniform magnetic field $B = 2.0\ \text{T}$ as shown in Fig. The ends of the rails are connected by a capacitor $C = 2\text{mF}$. A constant force $F = 1.2 \times 10^{-3}\ \text{N}$ is applied as shown. If the resistance of the rails is

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zero ohm, the acceleration of wire $PQ$ will be




**(A)** $0.1\ \text{ms}^{-2}$

**(B)** $0.5\ \text{ms}^{-2}$

**(C)** $1.0\ \text{ms}^{-2}$

**(D)** $2.0\ \text{ms}^{-2}$

**Q.3**

A rectangular, a square, a circular and an elliptical loop, all in the $(\text{x} - \text{y})$ plane, are moving out of a uniform magnetic field with a constant velocity, $\overrightarrow{\text{V}} = \text{v}\hat{\text{i}}$. The magnetic field is directed along the negative $z$ axis direction. The induced emf, during the passage of these loops, out of the field region, will not remain constant for

**(A)** the circular and the elliptical loops.

**(B)** only the elliptical loop.

**(C)** any of the four loops.

**(D)** the rectangular, circular and elliptical loops.

**Q.4**

A metal ring is held horizontally and bar magnet is dropped through the ring with its length along the axis of the ring. The acceleration of the falling magnet

**(A)** is equal to $\text{g}$

**(B)** is less than $g$

**(C)** is more than $\text{g}$

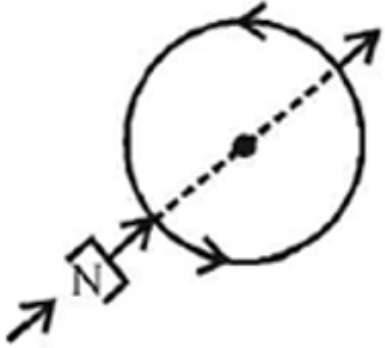
**(D)** depends on the diameter of ring and length of magnet

**Q.5**

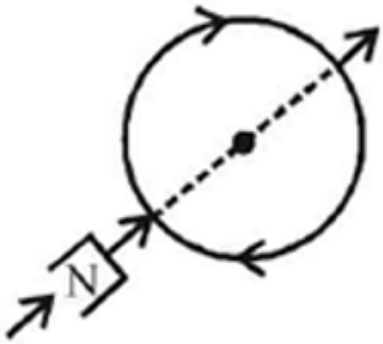
Which of the following figure correctly depicts the Lenz's law. The arrows show the movement of the labelled pole of a bar magnet into a closed circular loop and the arrows on the circle show the direction of the induced current

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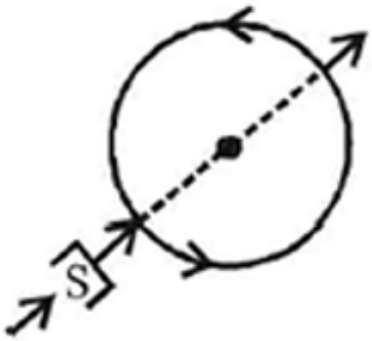
**(A)**



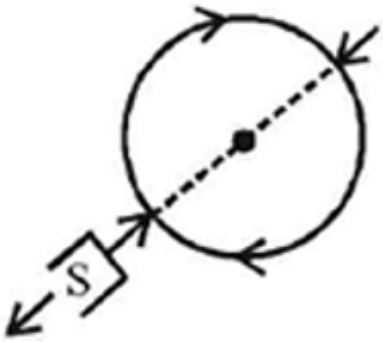

**(B)**




**(C)**




**(D)**

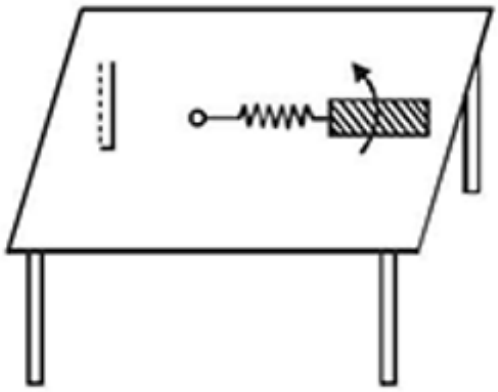



**Q.6**

A metallic rod of length ' $\ell$ ' is tied to a string of length $2\ell$ and made to rotate with angular speed $\omega$ on a horizontal table with one end of the string fixed. If there is a vertical magnetic field ' $B$ ' in the region, the e.m.f. induced

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across the ends of the rod is



**(A)** $\frac{2B\omega\ell^2}{2}$

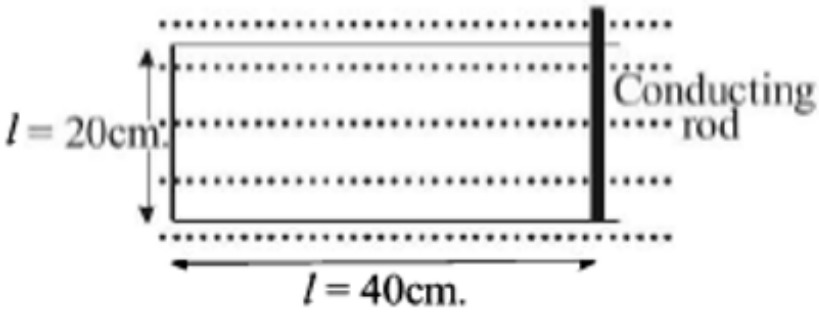
**(B)** $\frac{3B\omega\ell^2}{2}$

**(C)** $\frac{4B\omega\ell^2}{2}$

**(D)** $\frac{5B\omega\ell^2}{2}$

**Q.7**

Figure shows a conducting rod of negligible resistance that can slide on smooth U-shaped rail made of wire of resistance $1\Omega/\mathrm{m}$. Position of the conducting rod at $\mathrm{t} = 0$ is shown. A time $t$ dependent magnetic field $B = 2t$ tesla is switched on at $\mathbf{t} = 0$




At $\mathrm{t} = 0$, when the magnetic field is switched on, the conducting rod is moved to the left at constant speed $5\ \mathrm{cm}/\ \mathrm{s}$ by some external means. The rod moves perpendicular to the rails. At $\mathrm{t} = 2\ \mathrm{s}$, induced emf has magnitude

**(A)** $0.12\ \mathrm{V}$

**(B)** $0.08\ \mathrm{V}$

**(C)** $0.04\ \mathrm{V}$

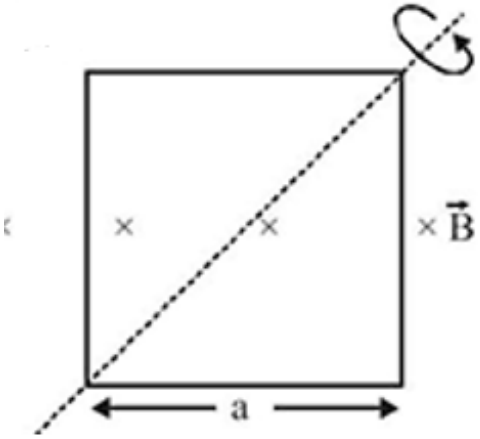
**(D)** $0.02\ \mathrm{V}$

**Q.8**

A square loop of side a is rotating about its diagonal with angular velocity $\omega$ in a perpendicular magnetic field $\overrightarrow{\mathrm{B}}$.

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It has 10 turns. The emf induced is




**(A)** $Ba^2 \sin \omega t$

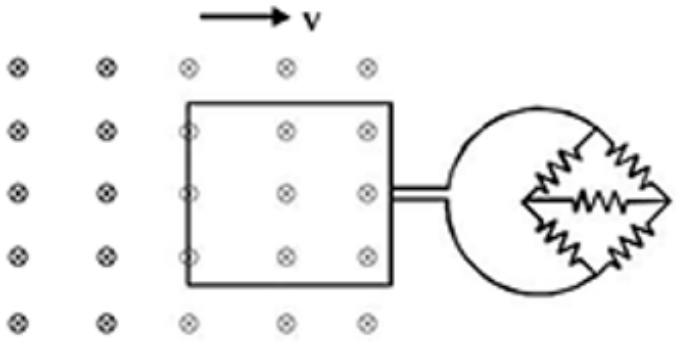
**(B)** $Ba^2 \cos \omega t$

**(C)** $5\sqrt{2}\, Ba^2$

**(D)** $10\, Ba^2 \omega \sin \omega t$

**Q.9**

A square metal loop of side $10\ cm$ and resistance $1\Omega$ is moved with a constant velocity partly inside a uniform magnetic field of $2 Wbm^{-2}$, directed into the paper, as shown in the figure. The loop is connected to a network of five resistors each of value $3\Omega$. If a steady current of $1\ mA$ flows in the loop, then the speed of the loop is




**(A)** $0.5 cms^{-1}$

**(B)** $1 cms^{-1}$

**(C)** $2 cms^{-1}$

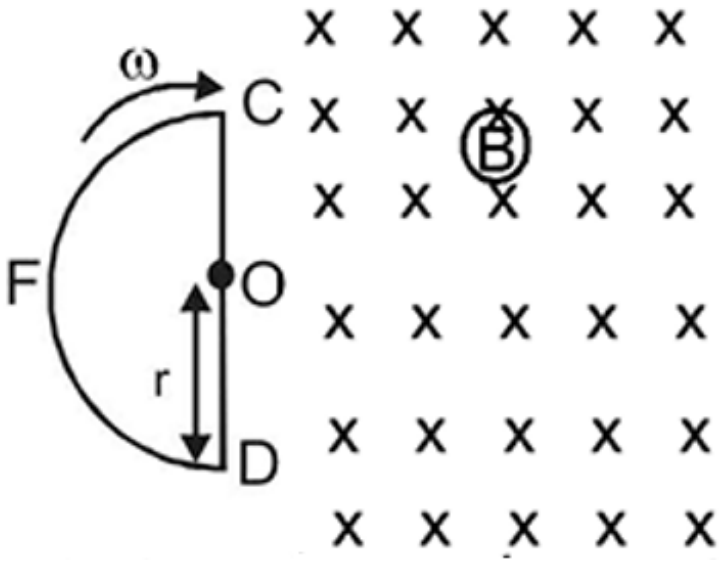
**(D)** $4 cms^{-1}$

**Q.10**

In fig, CODF is a semicircular loop of a conducting wire of resistance $R$ and radius $r$. It is placed in a uniform

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magnetic field $\mathrm{B}$, which is directed into the page (perpendicular to the plane of the loop).




The loop is rotated with a constant angular speed $\omega$ about an axis passing through the centre $\mathrm{O}$, and perpendicular to the page. Then the induced current in the wire loop is

**(A)** zero

**(B)** $\mathrm{Br}^2\omega/\mathrm{R}$

**(C)** $\mathrm{Br}^2\omega/2\mathrm{R}$

**(D)** $\mathrm{B}\pi\mathrm{r}^2\omega/\mathrm{R}$

**Q.11**

The magnetic flux through a circuit of resistance $\mathrm{R}$ changes by an amount $\Delta\phi$ in a time $\Delta t$. Then the total quantity of electric charge $Q$ that passes any point in the circuit during the time $\Delta t$ is represented by

**(A)** $\mathrm{Q} = \mathrm{R} \cdot \frac{\Delta\phi}{\Delta \mathrm{t}}$

**(B)** $\mathrm{Q} = \frac{1}{\mathrm{R}} \cdot \frac{\Delta\phi}{\Delta \mathrm{t}}$

**(C)** $\mathrm{Q} = \frac{\Delta\phi}{\mathrm{R}}$

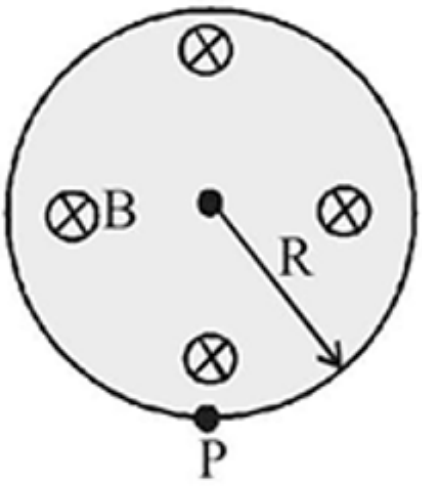
**(D)** $\mathrm{Q} = \frac{\Delta\phi}{\Delta \mathrm{t}}$

**Q.12**

A uniform magnetic field of induction $\mathrm{B}$ is confined to a cylindrical region of radius $\mathrm{R}$. The magnetic field is increasing at a constant rate of $\frac{\mathrm{dB}}{\mathrm{dt}}$ (tesla/second). An electron of charge

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e, placed at the point $\mathrm{P}$ on the periphery of the field experiences an acceleration.




**(A)** $\frac{1}{2}\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards left

**(B)** $\frac{1}{2}\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards right

**(C)** $\frac{\mathrm{eR}}{\mathrm{m}}\frac{\mathrm{dB}}{\mathrm{dt}}$ towards left

**(D)** zero

**Q.13**

A coil having $\mathrm{n}$ turns and resistance $\mathrm{R}\Omega$ is connected with a galvanometer of resistance $4\mathrm{R}\Omega$. This combination is moved in time $t$ seconds from a magnetic field $W_1$ weber to $\mathrm{W}_2$ weber. The induced current in the circuit is

**(A)** $-\frac{(\mathrm{W}_1-\mathrm{W}_2)}{\mathrm{Rnt}}$

**(B)** $-\frac{\mathrm{n}(\mathrm{W}_2-\mathrm{W}_1)}{5\mathrm{Rt}}$

**(C)** $-\frac{(\mathrm{W}_2-\mathrm{W}_1)}{5\mathrm{Rnt}}$

**(D)** $-\frac{\mathrm{n}(\mathrm{W}_2-\mathrm{W}_1)}{\mathrm{Rt}}$

**Q.14**

A metal rod of length 1 cuts across a uniform magnetic field $B$ with a velocity $v$. If the resistance of the circuit of which the rod forms a part is $\mathrm{r}$, then the force required to move the rod is

**(A)** $\frac{\mathrm{B}^2l^2v}{r}$

**(B)** $\frac{\mathrm{B}lv}{r}$

**(C)** $\frac{\mathrm{B}^2lv}{r}$

**(D)** $\frac{\mathrm{B}^2l^2v^2}{r}$

**Q.15**

A metal ring is held horizontally and bar magnet is dropped through the ring with its length along the axis of the ring. The acceleration of the falling magnet

**(A)** is equal to $\mathrm{g}$

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**(B)** is less than $g$

**(C)** is more than g

**(D)** depends on the diameter of ring and length of magnet

## Answers & Solutions

**Q.1 Answer:**

$2RBv$ and $Q$ is at higher potential.

**Solution:**

**As the ring falls with a velocity $v$ the decrease in area with time is**

$\frac{dA}{dt} = -(2R)v$

**$\therefore$ Induced emf,** $e = -\frac{d\phi}{dt} = -\frac{d}{dt}(BA) = -B\frac{dA}{dt} = 2RBv$

**From Lenz's law, the induced current in the ring must produce magnetic field in the upward direction. Hence $Q$ is at higher potential. Hence the correct choice is (d).**

**Q.2 Answer:**

$0.1\ \text{ms}^{-2}$

**Solution:**

**Let $v$ be the velocity of the wire at time $t$. The induced emf is $e = Blv$. The charge on the capacitor at time $t$ is**

$$q = Ce = CBlv$$

$$\therefore \quad \text{Current} \quad I = \frac{dq}{dt} = CBl\frac{dv}{dt} = CBla$$

**where $a =$ acceleration of the wire. The direction of $a$ (from Lenz's law) is to the left. Force on $PQ$ due to this current is**

$f = BIl = CB^2l^2a$ towards left.

**$\therefore$ Net force on wire is $F_{\text{net}} = F - f = F - CB^2l^2a$. The acceleration of wire by this force is**

$$a = \frac{F_{\text{net}}}{m} = \frac{F - CB^2l^2a}{m}$$

$$\Rightarrow \quad a = \frac{F}{m + CB^2l^2}$$

$$= \frac{1.2 \times 10^{-3}}{10^{-2} + 2 \times 10^{-3} \times (2.0)^2 \times (0.25)^2}$$

$$= 0.1\ \text{m s}^{-2}$$

**Q.3 Answer:**

the circular and the elliptical loops.

**Solution:**

**The induced emf will remain constant only in the case of rectangular and square loops. In case of the circular and the elliptical loops, the rate of change of area of the loops during their passage out of the**

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**field is not constant, hence induced emf will not remain constant for them.**

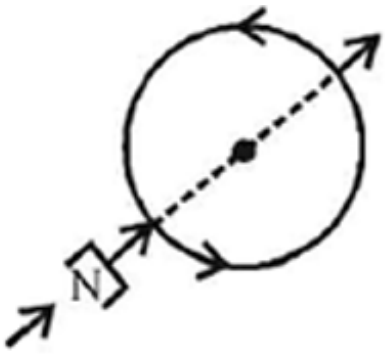
**Q.4 Answer:**

is less than $g$

**Solution:**

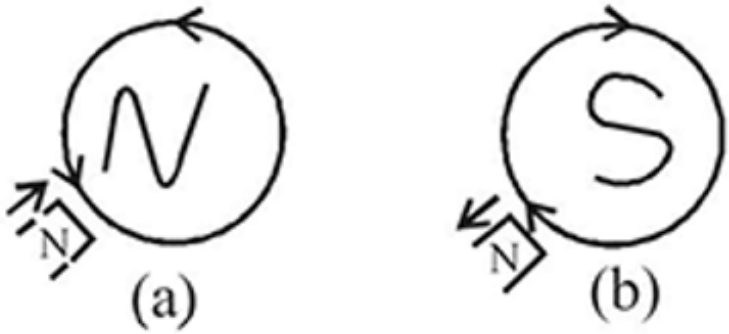
**Induced e.m.f. in the ring opposes the motion of the magnet.**

**Q.5 Answer:**




**Solution:**

**When a north pole of a bar magnet moves towards the coil, the induced current in the coil flows in a direction such that the coil presents its north pole to the bar magnet as shown in figure (a). Therefore, the induced current flows in the coil in the anticlockwise direction. When a north pole of a bar magnet moves away from the coil, the induced current in the coil flows in a direction such that the coil presents its south pole to the bar magnet as shown in figure (b).**




**Therefore induced current flows in the coil in the clockwise direction.**

**Q.6 Answer:**

$\frac{5B\omega\ell^2}{2}$

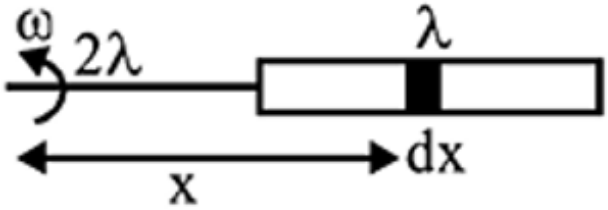
**Solution:**

**Here, induced e.m.f.**

$$e = \int_{2\ell}^{3\ell} (\omega x) B dx$$

$$= B\omega \frac{\left[(3\ell)^2 - (2\ell)^2\right]}{2} = \frac{5B\ell^2\omega}{2}$$

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**Q.7 Answer:**

$0.08\ \mathrm{V}$

**Solution:**

$\text{At } \mathrm{t}=2\ \mathrm{s},\ \mathrm{B}=4\ \mathrm{T}, \frac{\mathrm{dB}}{\mathrm{dt}}=\frac{2\ \mathrm{T}}{\mathrm{s}}$
$\mathrm{A}=20\times 30\ \mathrm{cm}^2$
$=600\times 10^{-4}\ \mathrm{m}^2, \frac{\mathrm{dA}}{\mathrm{dt}}=-(5\times 20)\mathrm{cm}^2/\mathrm{s}$
$=-100\times 10^{-4}\ \mathrm{m}^2/\mathrm{s}$
$\mathrm{E}=-\frac{\mathrm{d}\phi}{\mathrm{dt}}=-\left[\frac{\mathrm{d(BA)}}{\mathrm{dt}}\right]=-\left[\frac{\mathrm{BdA}}{\mathrm{dt}}+\frac{\mathrm{AdB}}{\mathrm{dt}}\right]$
$=-\left[4\times\left(-100\times 10^{-4}\right)+600\times 10^{-4}\times 2\right]$
$=-[-0.04+0.120]=-0.08\ \mathrm{V}$

**Q.8 Answer:**

$10\ \mathrm{Ba}^2\omega\sin\omega\mathrm{t}$

**Solution:**

$\phi=\mathrm{nBA}\cos\theta=10\ \mathrm{Ba}^2\cos\omega\mathrm{t}$
$\mathrm{e}=-\frac{\mathrm{d}\phi}{\mathrm{dt}}=-\frac{\mathrm{d}}{\mathrm{dt}}\left(10\mathrm{Ba}^2\cos\omega\mathrm{t}\right)=10\ \mathrm{Ba}^2\sin\omega\mathrm{t}(\omega)$

**Q.9 Answer:**

$2\mathrm{cms}^{-1}$

**Solution:**

$\varepsilon=\mathrm{B}\ell\mathrm{v}=2\times 10^{-1}\times\mathrm{v}=0.2\mathrm{v}$
$I=\frac{\varepsilon}{R}=10^{-3}\Rightarrow\frac{0.2v}{4}=10^{-3}$
**[Since effective resistance R of bridge is**
$\mathrm{R}=\frac{6\times 6}{6+6}=3\Omega$
**so total resistance** $=1+3=4\Omega$ **]** $\Rightarrow \mathrm{v}=2\ \mathrm{cm\ s}^{-1}$

**Q.10 Answer:**

$\mathrm{Br}^2\omega/2\mathrm{R}$

**Solution:**

**The area swept by radius OC in one half circle is** $\pi\mathrm{r}^2/2$**. The flux change in time** $\mathrm{T}/2$ **is thus** $\left(\pi\mathrm{r}^2\ \mathrm{B}/2\right)$**. The induced emf is then** $\mathrm{e}=\pi\mathrm{r}^2\ \mathrm{B}/\mathrm{T}=\mathrm{B}\omega\mathrm{r}^2/2\ \left[\because T=\frac{2\pi}{\omega}\right]$
**The induced current is then** $\mathrm{I}=\mathrm{e}/\mathrm{R}=\mathrm{B}\omega\mathrm{r}^2/2\mathrm{R}$

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**Q.11 Answer:**

$Q = \frac{\Delta\phi}{R}$

**Solution:**

$\frac{\Delta\phi}{\Delta t} = \varepsilon = iR \Rightarrow \Delta\phi = (i\Delta t)R = QR$
$\Rightarrow Q = \frac{\Delta\phi}{R}$

**Q.12 Answer:**

$\frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$ towards left

**Solution:**

**If we consider the cylindrical surface to be a ring of radius $R$, there will be an induced emf due to changing field.**

$$\int E d\vec{\ell} = -\frac{d\phi}{dt} = -A\frac{dB}{dt}$$

$\Rightarrow E(2\pi R) = -A\frac{dB}{dt} = -\pi R^2\frac{dB}{dt} \Rightarrow E = -\frac{R}{2}\frac{dB}{dt}$

**$\therefore$ Force on the electron**

$F = -Ee = \frac{eR}{2}\frac{dB}{dt}$

$\Rightarrow$ **acceleration** $= \frac{1}{2}\frac{eR}{m}\frac{dB}{dt}$

**As the field is increasing being directed inside the paper, hence there will be anticlockwise induced current (in order to oppose the cause) in the ring (assumed). Hence there will be a force towards left on the electrons.**

**Q.13 Answer:**

$-\frac{n(W_2 - W_1)}{5Rt}$

**Solution:**

$\frac{d\phi}{dt} = \frac{(W_2 - W_1)}{t} R_{tot} = (R + 4R)\Omega = 5R\Omega$
$i = -\frac{nd\varphi}{R_{tot}dt} = \frac{-n(W_2 - W_1)}{5Rt}$
($\because W_2 \& W_1$ **are magnetic flux)**

**Q.14 Answer:**

$\frac{B^2 l^2 v}{r}$

**Solution:**

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**Induced emf** $= B \vee l$

$$i = \frac{BVe}{r}$$

$$f_m = i\ell B$$

$$= \frac{Bv\ell \times l \times B}{r}$$

$$= \frac{B^2e^2V}{r}$$

**Q.15 Answer:**

is less than $g$

**Solution:**

**A metal ring is held horizontally and a bar magnet is dropped through the ring with its length along the axis of the ring. The magnet falls with an acceleration less than** g,

**As the magnet falls, the magnetic flux linked with the ring increases. This) induces emf in the ring which opposes the motion of the falling magnet, hence** $a < g$**.**

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